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Beyond Spatial Accuracy: Diagnosing Persistent Orientation Failures in Vision–Language Models

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Abstract

Evaluating AI/ML systems responsibly requires practices for testing, stress-testing, auditing, comparing, and interpreting specific capabilities—not only aggregate benchmark scores. This paper proposes such a practice for spatial reasoning in vision–language models (VLMs): a diagnostic audit of object-relative orientation, which aggregate accuracy can hide. On Visual Spatial Reasoning (VSR), moving from a compact 2.2B model to a 7B VLM substantially improves overall, depth, horizontal, and containment accuracy, but orientation changes only modestly; a large 2026 sparse-MoE model reproduces the pattern (79.5% / 65.7% overall/orientation), a newer flagship (GPT-5.6 Sol) does not reproduce the same relative orientation-weakness pattern (73.0% orientation), and a third external model (Gemini Spark) yields only a degenerate evaluation (chance-level, TRUE-defaulting; its redo reused the orientation answers, App. A). Except for structured prompting, trained conditions on both controlled models and the sparse-MoE model remain in a narrow 62–66% orientation band. Prompting, LoRA variants, visual-side adaptation, probes, and object-centric decomposition do not reliably close the gap, and a label-sensitivity analysis (an automated audit followed by two blind human re-audits; humans agree well on the failure taxonomy, but exclusion masks remain annotator-dependent and automated auditors vary substantially in calibration) shows that the pattern persists after excluding examples flagged as ambiguous.

Accuracy also fails to capture relational coherence. LM-side adaptation improves consistency over zero-shot; hard-negative training raises facing consistency from 66.0% to 77.7% with facing accuracy unchanged at 68.9% (pooled gain over LM-only n.s.). Cross-dataset evaluation on SITE shows VSR-trained adaptation does not transfer cleanly: neutral overall, significantly harming official spatial-relationship reasoning (75.1%→71.6%, $p = 0.004$). A pre-specified orientation-vocabulary slice, frozen before SITE evaluation, is analyzed as exploratory evidence. The results point to an orientation bottleneck that is persistent in the controlled model families and the sparse-MoE model but not universal (GPT-5.6 Sol does not reproduce the same relative pattern; Gemini Spark degenerates), in which orientation is not robustly decodable by the tested readouts and relational decisions are not always propagated coherently, while task-specific spatial adaptation can transfer poorly. Each component of the audit—the diagnostic ladder, the label-sensitivity check, the accuracy–coherence separation, and the cross-dataset transfer protocol—is specified operationally so it can be applied to other capabilities, benchmarks, and model families.

1. Introduction

This paper contributes an evaluation practice as much as a set of model results. We develop a protocol for testing, stress-testing, auditing, comparing, and interpreting one capability across vision–language models (VLMs), and instantiate it for object-relative orientation: facing, facing away, parallel, and perpendicular. This capability matters for robotics, embodied agents, navigation, and scene understanding, and is easy to hide behind an aggregate score: identifying the objects, inferring the relation, and answering complementary formulations are separable skills.

Prior benchmarks establish that grounded spatial reasoning remains difficult, but they leave open how to diagnose a relation-specific failure. VSR introduced a controlled natural-image probe and reported particularly poor object-orientation performance [7]; What'sUp and GSR-Bench similarly expose failures on basic grounded relations [6, 9]; and broader evaluations continue to identify orientation as a difficult factor [15]. We therefore ask three narrower questions: does the

VSR orientation weakness persist in modern generative VLMs and at much larger scale; can prompting, adaptation, or visual readouts remove it; and do accuracy, logical coherence, and cross-dataset transfer tell the same story?

The novelty is diagnostic rather than a claim that we discovered spatial difficulty. Our contribution is not any individual diagnostic, but a falsification protocol that holds the relation family and examples fixed while testing whether an apparent capability failure survives scaling and adaptation, label re-audit, representation readout, logical reformulation, and cross-dataset transfer. Prior work generally reports benchmark accuracy, studies a single intervention, or proposes a consistency objective; one controlled audit of the same family distinguishes a persistent relation-specific bottleneck from aggregate underperformance, annotation artifacts, or benchmark-specific specialization.

In our VSR experiments, the distinction between relation families is striking. Moving from SmolVLM2-2.2B-Instruct to the larger Qwen2-VL-7B-Instruct increases zero-shot overall accuracy from 74.0% to 80.9%; horizontal, depth, and containment rise by 6–15 pp. Orientation, by contrast, moves only from 62.8% to 63.5%. A large 2026 sparse-MoE model (MiMo-V2.5, 310B total / 15B active parameters) reproduces the same relation-specific pattern: 79.5% overall accuracy yet 65.7% on orientation, again inside the 62–66% band. Two further external zero-shot models bound this result: GPT-5.6 Sol does not reproduce the relative weakness—73.0% orientation, above its own depth (67.1%) and outside the band—while Gemini Spark's evaluation degenerates (51.5% overall; 98.6% TRUE answers in its first pass, a redo that reused the 137 orientation verdicts, App. A). The bottleneck is therefore model-family-dependent rather than universal, and one function of the protocol is precisely to detect such differences instead of assuming a failure generalizes. We then evaluate prompting, language-side adaptation, targeted data construction, visual-side adaptation, representation probing, and object-centric decomposition: structured prompting significantly hurts; general and targeted LoRA improve aggregate accuracy but do not remove the persistent orientation weakness; hard negatives improve targeted relational coherence without a significant orientation-accuracy gain; and the tested visual-side LoRA configurations do not outperform the language-side control.

As a sensitivity analysis, removing examples flagged by the automated audit raises orientation accuracy (65.7%→78.5% for 7B General LoRA, 66.4%→76.6% for hard-negative LoRA on the strictest 107-example subset) but does not eliminate error (2B structured shows no gain). The exclusion sets are annotator-dependent: two blind human re-audits agree strongly with each other on the failure taxonomy (Krippendorff's α=0.81) while their clean/ambiguous flags agree only moderately with the automated audit (κ=0.36 on the 48 shared persistent-failure cases); details are in the supplementary. Part of this gain is expected by construction, because the removed examples were drawn from persistent failures.

A second failure mode emerges from logical consistency. The zero-shot 7B model is consistent on only 36.9% of complementary facing pairs. LM-side LoRA substantially improves coherence, and hard negatives raise facing consistency to 77.7% with facing accuracy unchanged at 68.9%; the extra gain is not significant in the pooled strict-family comparison, yet accuracy stays fixed while consistency moves—the metrics capture different behavior.

Finally, we test cross-dataset transfer. The 7B model is strong on SITE's official spatial-relationship-reasoning category (75.1% raw; 59.2% chance-adjusted accuracy), yet the VSR-trained adapter does not generalize cleanly: it is statistically neutral overall (54.2%→53.8%, p = 0.66) and significantly degrades the official category (75.1%→71.6%, p = 0.004). A pre-specified, non-official orientation-vocabulary slice is much weaker in the aggregate (48.3% raw; 24.0% chance-adjusted) and shows no positive transfer, but—as we detail below—its composition limits what it can establish, and we therefore do not treat SITE as independent confirmation of the VSR orientation construct.

Our contributions are four reusable components of an evaluation protocol:

- A fine-grained diagnostic ladder. Across ten VSR conditions, two differently sized models, and three external zero-shot models, several spatial families improve substantially while object-relative orientation remains unusually resistant in the controlled families; one newer flagship does not reproduce the same relative orientation-weakness pattern and one external model's evaluation degenerates under the protocol, bounding rather than weakening the claim. The ladder—scaling, prompting, language-side adaptation, targeted data construction, visual-side adaptation, representation probing, and explicit decomposition—is defined operationally so it transfers to other benchmarks and relation families.
- A robustness and sensitivity audit. A label-sensitivity analysis (automated labels with two independent human re-audits), frozen/object-grounded probes, visual-side adaptation, and object-centric decomposition constrain purely annotation-based or single-component explanations. The audit practice

(flagged-example exclusion sets, decodability checks against majority baselines) applies to any benchmark whose labels or examples may be ambiguous.

- An accuracy–coherence separation protocol. Complementary-relation tests on VSR's strict relation pairs show that, for the tested conditions, relational consistency can change substantially while task accuracy remains fixed. The protocol measures consistency against verdict-pattern base rates, so it can be applied wherever complementary formulations of the same relation exist.
- A cross-dataset transfer protocol. SITE shows strong official spatial-relationship performance, while VSR-trained LoRA transfers poorly and significantly harms the official SITE category; the orientation-vocabulary slice is analyzed as exploratory evidence. The protocol (pre-specified subsets, chance-adjusted accuracy, paired transfer tests) is model- and benchmark-agnostic and is reported with negative as well as positive transfer.

The protocol is deliberately modular: each component can be applied independently to other capabilities, benchmarks, and model families. The supplementary specifies the prompts, subsets, decision rules, and released artifacts needed to reproduce the audit.

2. Related Work

Benchmarks for grounded spatial reasoning. VSR introduced a controlled natural-image benchmark with 66 spatial relations and identified facing and parallel as difficult relations for the VLMs studied at the time [7]. What'sUp controls object identity while varying spatial configuration and reports a substantial human–model gap [6]; GSR-Bench broadens grounded evaluation across models and scales [9]. SITE extends coverage to single-image, multi-image, and video inputs and organizes tasks by dimensions such as orientation and intrinsic/extrinsic relations [15]. SpatialScore and Mind the Gap further show that spatial competence remains fragmented [10, 16]. We use VSR not to re-establish that spatial reasoning is hard, but to turn its orientation signal into a controlled diagnostic target and use SITE to test transfer.

Diagnosing and improving spatial failures. Several lines of work move beyond aggregate accuracy. AdaptVis relates successful VSR reasoning to attention on relevant object locations [2], while CREG shows that attribution evidence in Qwen2-VL can reflect image layout rather than the queried relation [11]. When More Is Less finds that spatial cues, commonsense information, and reasoning instructions can help or hurt depending on the model and setup [1]. Other work probes visual-concept decodability [13], separates perception from reasoning in spatial planning [17], or supplies geometric inputs for 3D spatial understanding [3]. These motivate our combination of frozen/readout diagnostics with adaptation and object-centric decomposition, but none audits one relation family across accuracy, coherence, label sensitivity, and transfer.

Taken together, prior work establishes the problem and offers several useful diagnostic lenses, but leaves a methodological gap: benchmark accuracy, representation access, logical consistency, annotation sensitivity, and transfer are usually reported separately. We close that gap with a single controlled diagnostic ladder, with analysis choices and external-evaluation subsets frozen where specified, applied to the same relation family. Our claim is consequently not to discover spatial difficulty, but to make a persistent orientation weakness measurable, falsifiable, and reusable as an evaluation protocol.

Adaptation, specialization, and transfer. Parameter-efficient adaptation such as LoRA [5] can improve a benchmark without improving the underlying capability; we therefore compare general, targeted, hard-negative, projector, and vision-plus-projector updates and evaluate the strongest adapter on an untouched benchmark. VSR adaptation is neutral overall on SITE, does not improve the orientation subset, and significantly degrades the official spatial-relationship category—negative transfer treated as an evaluation result rather than an omitted outcome.

Representation probing and logical consistency. Probes can test whether a signal is accessible from frozen representations, but probe capacity can itself learn the task, requiring control baselines and cautious interpretation [4]. Consistency-oriented objectives such as SAGE use geometric and linguistic dualities to train spatial reasoning [8]. We instead use probing and complementary-pair consistency as independent diagnostics: the former asks what the tested readouts can decode, while the latter asks whether answers to logically linked statements agree, so accuracy and coherence can diverge as diagnostic axes.

3. Experimental Setup

3.1. Benchmarks

VSR. We use VSR [7] as the primary diagnostic benchmark. The random-split test set contains 2,195 examples. The benchmark defines 64 relation labels; 61 appear in the evaluated test split. The orientation family contains facing, facing away from, parallel to, and perpendicular to, totaling 137 test examples. All main VSR results use the same held-out test set and deterministic True/False parsing.

SITE. We use SITE [15] for cross-dataset transfer evaluation only and do not train on SITE. The image evaluation contains 2,591 examples: 1,368 single-image and 1,223 multi-image examples. Our primary SITE subset is the official spatial-relationship-reasoning category ($n = 993$). We additionally report a transparently defined, non-official keyword-derived orientation subset ($n = 2{,}024$), analyzed as exploratory evidence rather than as direct replication of the VSR construct. SITE results use the benchmark's native multiple-choice format and chance-adjusted accuracy (CAA) in addition to raw accuracy.

3.2. Models and Conditions

Our compact model is SmolVLM2-2.2B-Instruct, and our larger controlled-study model is Qwen2-VL-7B-Instruct [14]. We retain Qwen2-VL-7B as the controlled anchor because it supports the complete local ladder—the same base model across prompting, LoRA, visual-side adaptation, probing, and decomposition—rather than mixing intervention effects with an uncontrolled architecture change. We evaluate ten canonical VSR conditions: 2B zero-shot, 2B structured prompting, 2B General LoRA, 2B Targeted LoRA, 7B zero-shot, 7B General LoRA, 7B Targeted LoRA, 7B Hard-Neg LoRA, 7B Projector LoRA, and 7B Vision+Projector LoRA. LoRA [5] is used for all parameter-efficient adaptation conditions. We additionally evaluate a large zero-shot sparse-MoE model, MiMo-V2.5 [12] (310B total / 15B activated parameters, dedicated 729M-parameter vision encoder, 2026), served through its public API with the identical frozen prompt, thinking disabled, and greedy decoding (App. A). Its role is external validation of the diagnosed pattern; it is not fine-tuned. Two additional external zero-shot models used the same frozen prompt, the same parser, and the same invalid-counted-wrong policy: GPT-5.6 Sol (OpenAI, per-item evaluation, 2026-08-12; App. A) and Gemini Spark (Google, per-item run run_gemini_spark_vsr2195_v2_001). Gemini's first pass defaulted to TRUE on 98.6% of items; its redo reused the 137 orientation answers (id-level verified), so its Table 1 row is a degenerate-evaluation finding, not a capability estimate (App. A).

General LoRA uses a proportional sample across relation families; Targeted LoRA concentrates training on weak orientation/depth/horizontal families with rehearsal of the rest; Hard-Neg LoRA adds audited complementary orientation examples, primarily facing/facing-away inversions; vision-side conditions adapt the multimodal projector or upper vision blocks with it.

3.3. Metrics and Statistical Tests

For VSR we report accuracy overall and by spatial family. For SITE we report raw accuracy and CAA. Paired condition comparisons use McNemar tests when applicable. Clean-label orientation analysis reports the full orientation set and progressively stricter audited subsets. Logical consistency is evaluated by constructing complementary statements for strict relation pairs (left/right, front/behind, facing/facing-away) and testing whether the model assigns opposite verdicts as required. Parallel/perpendicular is treated separately as a soft complement because both relations can be false for oblique objects.

4. Main Results

4.1. Orientation Remains Weak Across Two Differently Sized VLMs

Table 1 shows the central pattern. Moving from the 2B model to the 7B model lifts zero-shot overall accuracy by 6.9 pp, horizontal by 14.6 pp, depth by 6.3 pp, and containment by 5.9 pp, but orientation by only 0.7 pp. General 7B LoRA further raises overall accuracy to 84.7%, yet orientation reaches only 65.7%. Across trained conditions other than structured prompting, orientation remains in a narrow 62–66% band. The large sparse-MoE zero-shot model (MiMo-V2.5, evaluated identically, App. A) falls in the same band: 79.5% overall and 65.7% orientation, with the orientation family again its weakest relation family (per-family table in the supplementary). The relation-specific pattern is therefore not an artifact of the 2B/7B scale range. Two further external zero-shot models bound the pattern: GPT-5.6 Sol does not reproduce the relative weakness (73.0% vs. its own 67.1% depth; CI $[65.0, 79.7]$ overlaps the band), while Gemini Spark's evaluation degenerates (51.5% overall; App. A). The MiMo-V2.5 row is also robust to output-format parsing: orientation is unchanged parseable-only (65.7%), and parseable-only overall is 82.1% vs. 79.5% counting the 69 nonconforming outputs wrong (App. G).

A paired bootstrap over the 2,195 shared examples (10,000 resamples) confirms the horizontal-vs-orientation scaling contrast (+13.9 pp; 95% CI $[+2.9, +24.8]$; positive in 99.3% of resamples); depth (+5.5 pp) and containment (+5.1 pp) are directionally positive but not individually significant (App. D).

Structured prompting is a negative result: overall accuracy falls from 74.0% to 68.3% at 2B, while ori-

Table 1. Canonical VSR test accuracy (%). Orientation improves little with model size or adaptation compared with several other relation families.

Condition	Overall	Orientation	Depth	Horizontal	Containment	Topology/contact
2B zero-shot	74.0	62.8	68.9	70.2	83.4	80.5
2B structured	68.3	53.3	64.9	66.7	78.7	71.7
2B General LoRA	76.6	62.0	71.1	74.3	87.6	81.0
2B Targeted LoRA	76.5	64.2	70.8	75.1	87.6	81.4
7B zero-shot	80.9	63.5	75.2	84.8	89.3	80.3
7B General LoRA	84.7	65.7	82.3	87.4	92.9	84.4
7B Targeted LoRA	83.9	64.2	81.7	87.1	91.7	85.3
7B Hard-Neg LoRA	84.3	66.4	80.7	87.1	89.9	84.6
7B Projector LoRA	82.9	64.2	78.6	87.1	89.3	85.3
7B Vision+Projector LoRA	83.1	64.2	78.0	87.1	89.9	84.4
MiMo-V2.5 zero-shot	79.5	65.7	74.8	81.9	87.1	84.6
GPT-5.6 Sol zero-shot	75.4	73.0	67.1	76.9	82.8	77.8
Gemini Spark zero-shot†	51.5	57.7	55.0	53.2	50.3	47.5

†Gemini Spark's evaluation is degenerate (98.6% TRUE in pass 1; redo reused the orientation answers, id-level verified); the row is a degenerate-evaluation finding, not a capability estimate (App. A).

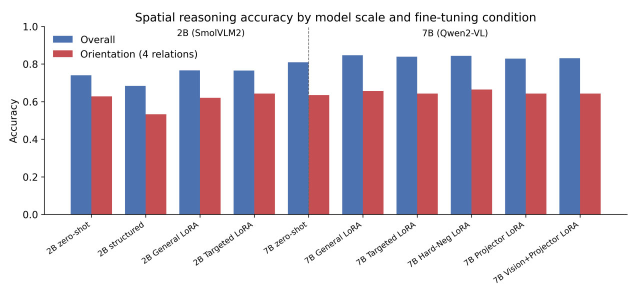

Figure 1. Overall versus orientation performance across the two VLMs and adaptation conditions.

entation falls from 62.8% to 53.3%. A paired exact McNemar test on the full test set confirms that the structured prompt is worse than the minimal baseline ($p \approx 2.2 \times 10^{-7}$). Consistent with recent VSR analyses where additional cues or reasoning instructions help or hurt depending on setup [1], this is specific to the tested 2B condition and prompt.

4.2. Clean-Label Sensitivity Analysis

An automated audit (LLM-generated labels; frozen v1) flagged ambiguous or questionable orientation examples; we repeat evaluation on progressively stricter subsets as a sensitivity analysis (full table in the supplementary). The best full-set orientation result is 66.4%; on the automated strictest 107-example subset, the best result rises to 78.5% for 7B General LoRA. Because auditing changes the sample, these are not matched comparisons across families; two blind human re-audits (binary flag on all 137; eight-class taxonomy on the 48 persistent failures) yield three distinct analyses (App. F).

Taxonomy-derived strict masks. Each rater's strict mask raises orientation accuracy without eliminating error—automated 78.5% (n=107), human A 77.6% (n=107), human B 78.8% (n=104)—and 2B structured prompting still shows no gain under any of the three masks (53.3%, 51.4%, 51.0%).

Binary clean/ambiguous masks. The two humans agree substantially on the 137-example binary flag (κ=0.645, 82.5% agreement) but only moderately with the automated audit (κ=0.36 on the 48 cases rated by both sources; the automated audit never flagged the remaining 89), and the taxonomy agrees only weakly with the automated labels (α=0.14). Under the human binary masks the gains largely vanish (best 66.7%/66.2% vs. 66.4% full-set), tying gains to taxonomy-derived masks; audit reliability is itself model-dependent—GPT-5.6 Sol (κ=0.34) and Gemini Spark (κ=0.03, 126/137 clean) agree with humans only weakly—so LLM-generated ambiguity labels are sensitivity tools, not ground truth (App. F).

Consensus analysis. Under the consensus union mask (n=62), trained conditions again improve (7B General LoRA 77.4%) but 7B zero-shot does not (58.1%), and 2B structured's no-gain pattern does not survive (69.3%), consistent with annotator dependence.

The robust conclusion is narrower: taxonomy-derived strict masks raise orientation scores but do not eliminate error across conditions; the audit is exploratory qualitative evidence, not a ruling on label noise (frozen v1 and both human tables in the supplementary).

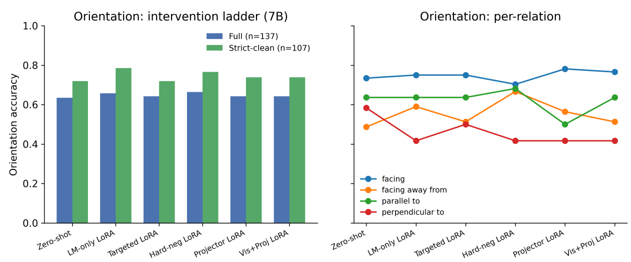

Figure 2. Orientation performance across 7B interventions, including clean-label analysis and per-relation behavior.

5. Why Does Orientation Remain Difficult?

5.1. Frozen Representation Probes

We probe frozen Qwen2-VL-7B-Instruct visual representations (no fine-tuning). Mean-pooled ViT and post-merger features support only weak relation decoding for facing versus facing-away, parallel versus perpendicular, and a four-way orientation task. Linear and MLP probes generally remain near majority baselines; preserving patch-level structure yields a suggestive point estimate for parallel/perpendicular (patch-vote 73.5% vs. 64.7% majority), but that test set is small (n=34) and the CI overlaps the majority baseline.

To reduce the confound that global features infer relations without identifying objects, we extract subject/reference regions and construct grounded features from subject, object, difference, and elementwise-product representations; object-box conditioning with the model's own grounding predictions and mean-pooled region features does not recover robust orientation decodability. Following standard cautions about diagnostic classifiers [4], we phrase these results as decodability under the tested readouts, not as proof that the information is absent from the network. Probing decodability of visual concepts has precedent in driving-oriented VLMs [13] and attribution analyses of this model family [11]; ours targets VSR's orientation family with pooled, patch-vote, and object-grounded readouts over the model's exact conditioning features, paired with adaptation and consistency interventions. Detailed probe tables and plots are in the supplementary.

5.2. Visual-Side Adaptation and Object-Centric Decomposition

If the frozen representation were the sole limiting factor, adapting visual or projector components might recover orientation. For the trained checkpoints here it did not: projector and vision+projector LoRA both produced 64.2% orientation, below the 65.7% LM-only control, with overall nominally lower (82.9%/83.1% vs. 84.7%; unadjusted p=0.0043/0.0122, not surviving blanket adjustment and not load-bearing, App. D).

We also test a box-based two-stage decomposition. Its four-way relation module classifies facing/facing-away/parallel/perpendicular against ground-truth relation labels at 44–46% cross-validation accuracy (49.9% majority baseline; uniform four-class chance 25%), so the tested relation classifier does not outperform the empirical majority baseline. Consistent with that, the pipeline's verdicts match the control's correctness on only 55.1–58.3% of the 127 boxed statements (per-relation agreement 25.0–69.4%, weakest for parallel); in the committed analysis (10 no-box statements scored wrong) they diverge from the control on 61–65 of 137, with the control correct and the pipeline wrong in 44–47 of them (App. D). These are agreement scores, not ground-truth accuracies: the committed script scored the pipeline against the control's verdicts, so the decomposition is reported as an agreement diagnostic that fails to reproduce the model's behavior, not as a direct accuracy comparison (ground-truth scoring via the released script, App. G). Two-dimensional box geometry is structurally insufficient for intrinsic facing direction, and the tested region representations do not close the gap.

5.3. Logical Consistency Reveals a Distinct Failure Mode

We construct complementary statements for strict relation families. A coherent model should return opposite truth values for left/right, front/behind, and facing/facing-away formulations of the same object pair. Complementary-pair consistency has prior literature, including consistency-oriented training [8]; our diagnostic below uses strict pairs with verdict-pattern and base-rate analysis. The zero-shot 7B model is consistent on only 58.0% of left/right pairs, 57.0% of front/behind pairs, and 36.9% of facing pairs. The large sparse-MoE model does not repair coherence: 42.7% facing-pair consistency vs. 36.9% (44.0% on the 100 parseable pairs, three excluded for non-conforming outputs), with the same False-defaulting pattern (0 both-True, 56.0% both-False; verdict table in the supplementary). Scale alone therefore does not resolve the coherence failure mode. The two external models diverge: GPT-5.6 Sol reaches 83.3% complementary facing verdicts (no both-True, n=102; 54.5% expected under independence), far above the controlled families, while Gemini Spark's 60.2% with a zero flip-True rate and 100% complementary front/behind pairs are treated as degenerate, not capability estimates (App. A). The verdict patterns show a False-defaulting

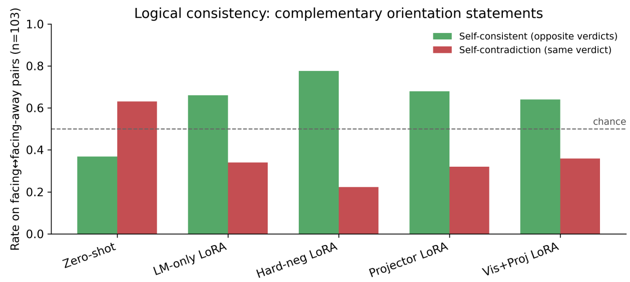

Figure 3. Self-consistency on complementary facing/facing-away relations across 7B conditions.

bias (both-False 63.1%, both-True 0%, above the 30.2% expected under verdict independence; verdict table in the supplementary).

LM-only LoRA raises facing consistency to 66.0% and front/behind to 70.7% (pooled strict-family significantly better than zero-shot, $p < 0.0001$); hard negatives raise facing consistency to 77.7% with facing accuracy identical at 68.9% (pooled gain over LM-only n.s., $p = 0.29$). The key result is separability: identical facing accuracy coexists with substantially different consistency. Training changes the failure mode rather than eliminating it: both-False facing verdicts drop from 63.1% to 9.7% while both-True contradictions rise from 0% to 24.3%, a disclosed trade-off.

6. Cross-Dataset Transfer to SITE

We evaluate the frozen 7B base model and the VSR-trained 7B General LoRA adapter on the same 2,591 SITE image examples using SITE's native multiple-choice protocol [15], with no SITE training. Because the pre-specified decision rule resolved to branch 2, this adapter run is a post-protocol transfer check rather than a pre-specified step (protocol and outcome in the supplementary). The model is strong on the official spatial-relationship-reasoning category: 75.1% raw accuracy and 59.2% CAA. Applying the VSR-trained adapter does not transfer cleanly: it is statistically neutral overall (54.2%→53.8%, $p = 0.66$) and significantly degrades the official category (75.1%→71.6%, $p = 0.004$; 52 examples fixed vs. 87 broken). VSR gains therefore transfer poorly and can induce negative transfer; we do not equate benchmark-specific fine-tuning gains with general spatial-reasoning improvement.

The pre-specified orientation-vocabulary slice is much weaker in the aggregate (48.3% raw / 24.0% CAA vs. 75.1% / 59.2% official) with no positive transfer (48.3%→48.7%, $p = 0.70$). Because the slice is keyword-derived and heterogeneous (43% of its examples come from multi-view/cross-image reasoning, only

Table 2. SITE cross-dataset transfer (images). Entries are raw accuracy / CAA (%).

Subset	Zero-shot	VSR-LoRA	Δ
All images	54.2 / 31.1	53.8 / 30.5	−0.4
Official spatial-rel.	75.1 / 59.2	71.6 / 53.4	−3.5
Orient. heuristic	48.3 / 24.0	48.7 / 24.5	+0.4

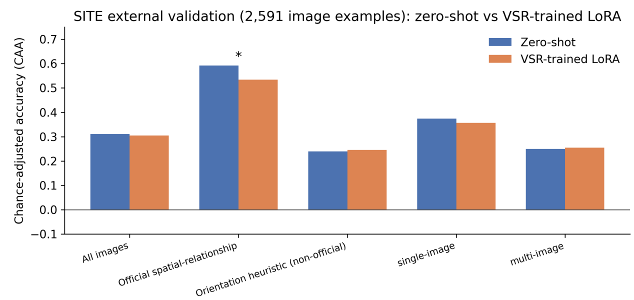

Figure 4. SITE CAA for zero-shot and VSR-trained LoRA. The official spatial-relationship category degrades significantly under transfer.

24% from the spatial-relationship category), we treat it as exploratory evidence rather than replication of the VSR construct. A composition-confound analysis (supplementary §E) supports this caution: after controlling for official category, source dataset, modality, and option count, the orientation flag is not significantly associated with correctness (OR 0.84, 95% CI $[0.60, 1.16]$, $p = 0.29$), though descriptively 7.5pp lower within the official category itself (71.3% vs. 78.8%, n=488/505). A post-hoc high-precision subset restricted to VSR-construct terms (facing, facing away, parallel, perpendicular; n=177) yields 44.1% raw (95% CI $[37.0, 51.4]$; CAA −2.3%) and is reported as exploratory only. We therefore do not treat SITE as independent confirmation of the VSR orientation bottleneck; the transfer results stand on the official category, where the model is strong and the adapter demonstrably does harm.

7. Discussion

Orientation is not spatial reasoning broadly. The strongest conclusion is deliberately narrow. The 7B model is strong on several VSR families and SITE's official spatial-relationship category. The persistent weakness concerns the object-relative orientation family in VSR; on SITE, the orientation-vocabulary slice is analyzed as exploratory evidence rather than as independent confirmation. This framing is consistent with the original VSR observation [7] while extending it to modern generative VLMs and a broader intervention analysis. The weakness is also model-family-dependent: the controlled families and the sparse-MoE model persist in it, GPT-5.6 Sol does not reproduce it, and Gemini Spark's evaluation degenerates under the protocol; a diagnostic that only found the failure would be uninformative, so this distinguishability is what makes the protocol falsifiable.

Decodability and coherence both matter, as diagnostics. The probes show that orientation is not robustly decodable under the tested frozen-feature readouts, while consistency experiments expose large logical failures even for relations answered above chance. Neither alone identifies a causal mechanism; together they support a mixed account in which orientation information is hard to access under the tested readouts and relational decisions are not always propagated coherently.

Task adaptation can specialize rather than generalize. General LoRA improves VSR overall accuracy substantially, but its SITE transfer is neutral overall and significantly negative on the official subset. Hard negatives improve a targeted coherence behavior with relation-specific tradeoffs. Benchmark-specific fine-tuning gains should therefore not be equated with general spatial-reasoning improvement.

8. Limitations

Statistical power is limited at the scale of the orientation diagnostic. The VSR orientation family contains only 137 test examples, and the smallest per-relation subset, perpendicular, has n=12; perpendicular results are descriptive only. The Wilson 95% confidence intervals on the family-level point estimates are correspondingly wide—62.8% [54, 70], 63.5% [55, 71], 65.7% [57, 73], and 66.4% [58, 74] for 2B zero-shot, 7B zero-shot, 7B General LoRA, and 7B Hard-Neg LoRA (n=137 each)—so the trained conditions' 62–66% band should be read as a set of statistically overlapping point estimates rather than a fine-grained ranking. Several smaller point estimates have confidence intervals that include the majority baseline or chance: the patch-vote parallel/perpendicular probe (73.5%, n=34; 95% CI [57, 85] vs. the 64.7% majority), the perpendicular subset for the trained 7B conditions (41.7%, n=12; [19, 68] vs. 50%), and the 7B zero-shot facing-away subset (48.7%, n=39; [34, 64]). Paired McNemar tests on orientation rest on small discordant-pair counts: the non-significant projector and vision+projector comparisons (p=0.86, p=0.85) indicate no detectable effect, not equivalence; conclusions rely on the full-set paired tests and the Holm-adjusted battery (supplementary).

The clean-label audit's exclusion sets originate from an automated LLM audit: two independent human re-audits reached only moderate per-item agreement with it (κ=0.36 on the clean/ambiguous flag; Krippendorff's α=0.14 on the eight-class taxonomy) while agreeing strongly with each other on the failure taxonomy (α=0.81); both reproduced the qualitative conclusions under their own taxonomy-derived strict masks, so the sets are annotator-dependent (tables versioned: frozen v1 plus both human re-audits). The SITE orientation slice is keyword-derived; a composition-confound analysis shows its aggregate score is largely explained by task composition, so SITE is not independent confirmation of the VSR bottleneck. Our probing methods test accessible decodability under specific linear/MLP and pooling choices; failure to decode does not prove information is absent. Visual-side experiments use LoRA rather than full retraining; the three model families form a cross-model comparison, not a controlled scale experiment. All trained conditions were single-checkpoint runs: paired McNemar tests quantify disagreement on the fixed test examples, not training-run variability. A completed multi-seed extension (3 seeds for 7B General LoRA, 2 each for Targeted and Hard-Neg; identical recipe, seed-only change) shows overall accuracy is stable across seeds (±0.1–0.4 pp SD) while orientation accuracy varies ±0.5–3.1 pp; every between-condition orientation delta falls inside the seed-to-seed SD (App. D). External API-served models (MiMo-V2.5, GPT-5.6 Sol, Gemini Spark; all evaluated on 2026-08-12 with the identical frozen prompt) can change without notice, so their reproducibility depends on the committed per-item predictions and serving-snapshot records (App. A); Gemini's evaluation is degenerate (98.6% TRUE in pass 1; redo reused 1,315/2,195 answers including all 137 orientation verdicts, id-level verified), so its row and coherence numbers are a documented degenerate-evaluation outcome, not a capability estimate. The SITE evaluation is image-only; SITE video reasoning remains future work. Finally, several interventions are evaluated as negative results; lack of improvement holds for the tested configurations, not as impossibility claims.

9. Conclusion

Aggregate accuracy hides capability-specific failures: adaptation improves several spatial families while leaving orientation comparatively resistant, and external models bound the pattern's universality. Auditing, probing, and decomposition constrain simple explanations; accuracy and coherence are distinct axes; benchmark-specific adaptation can negatively transfer. Evaluation practice should report relation-specific performance, transfer, and logical coherence rather than aggregate accuracy alone.

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